

Packet Tracer - Compare Layer 2 and Layer 3 Devices

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Part 1: Compare Layer 2 and Layer 3 Switches

Step 1: Explore the Physical Workspace

Each individual switch has how many physical switchports?

ASw-1 (2960-24TT): 26 switchports (FastEthernet0/1-24 and GigabitEthernet0/1-2). **D1 (3650-24PS): 28 switchports** (GigabitEthernet1/0/1-24 and GigabitEthernet1/1/1-4).

How many Fast Ethernet and Gigabit Ethernet switchports does each switch have?

Switch	Fast Ethernet	Gigabit Ethernet	Total
ASw-1 (2960-24TT)	24	2	26
D1 (3650-24PS)	0	28	28

List the transmission speed of the Fast Ethernet and Gigabit Ethernet switchports on each switch.

Fast Ethernet ports run at 100 Mbps and Gigabit Ethernet ports run at 1000 Mbps (1 Gbps). ASw-1 has 24 ports at 100 Mbps and 2 ports at 1 Gbps. All 28 ports on D1 run at 1 Gbps.

Are either of the two switches modular in design?

Neither switch is modular in its switchports. On the Physical tab, the ASw-1 (2960) Modules list is empty because it is a fixed-configuration switch with nothing to add. D1 (3650) has 24 built-in access ports that cannot be changed; its Modules list only offers an AC power supply, a power cover plate, and SFP transceivers (GLC-LH-SMD, GLC-T, GLC-TE) for the uplink ports.

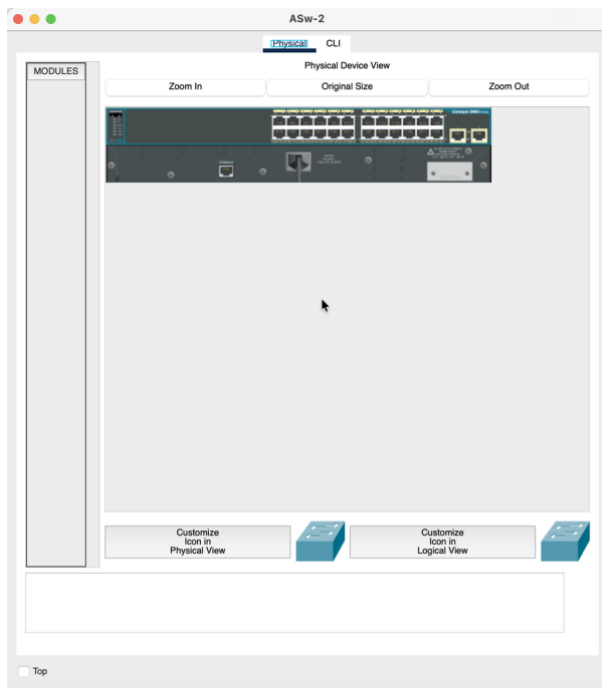


Figure 1a. Physical tab of the 2960-24TT switch (ASw-2 shown; same model as ASw-1) - no modules available

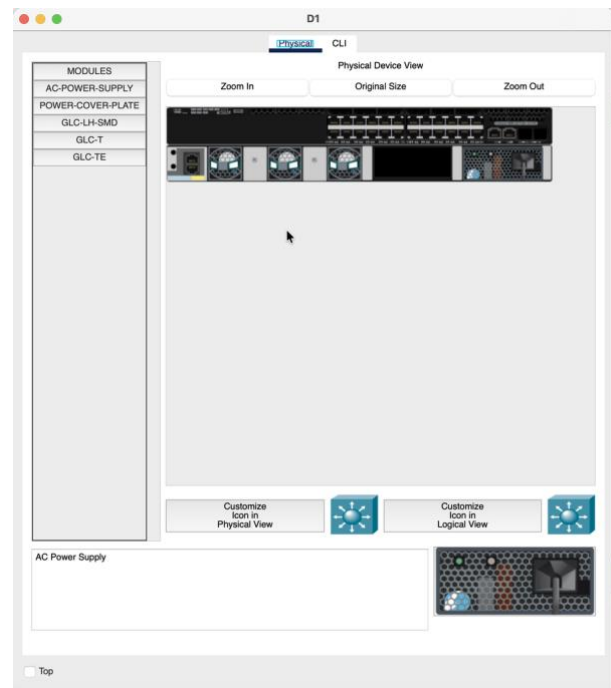


Figure 1b. Physical tab of D1 (3650-24PS) - only power supply and SFP modules

Step 2: In the CLI tab

What is the difference between a Layer 2 switch and a Layer 3 switch?

A Layer 2 switch forwards frames within a network using MAC addresses. It cannot route between networks. A Layer 3 switch does everything a Layer 2 switch does and can also route IP packets between different networks. It keeps a routing table, can run routing protocols when ip routing is enabled, and can have routed ports (no switchport) with IP addresses like a router.

What is the difference between a switch's physical interface and the VLAN interface?

A physical interface is an actual port on the switch (for example GigabitEthernet1/1/1). A VLAN interface (SVI) is a virtual Layer 3 interface that represents a VLAN and is not tied to one physical port. On the Layer 2 switch ASw-1 the IP address is on Vlan1 (192.168.10.2) and is used only for management. On the Layer 3 switch D1 the IP addresses are on physical routed ports (Gi1/1/1 and Gi1/1/2).

At which layers do 2960 and 3650 switches operate?

The 2960 operates at Layer 2 only. The 3650 operates at Layer 2 and Layer 3.

Do you notice any differences between D1 and ASw-1 in show run?

Yes. D1 has ip routing and routed ports, while ASw-1 does not.

	ASw-1 (2960, Layer 2)	D1 (3650, Layer 3)
ip routing	Not present	Present
Physical ports	Layer 2 only, no IP address	Gi1/1/1 and Gi1/1/2 use no switchport with IP addresses
Vlan1	IP 192.168.10.2/24, interface up	no ip address, shutdown
Default gateway	ip default-gateway 192.168.10.1	Not needed
IOS version	12.2	16.3.2

Excerpts of show run (unrelated lines omitted). ASw-1 has no ip routing command.

```
D1#show run
...
no ip cef
ip routing
...
interface GigabitEthernet1/1/1
  no switchport
  ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet1/1/2
  no switchport
  ip address 192.168.11.1 255.255.255.0
!
interface Vlan1
  no ip address
  shutdown

ASw-1#show run
...
interface GigabitEthernet0/2
!
interface Vlan1
  ip address 192.168.10.2 255.255.255.0
!
ip default-gateway 192.168.10.1
```

Why do you think the show ip route command does not work on ASw-1 but works on D1?

ASw-1 is a Layer 2 switch. It forwards frames by MAC address, has no routing function, and does not build a routing table, so the command is not accepted. D1 is a Layer 3 switch with ip routing enabled, so it has a routing table and the command works.

```

ASw-1#show ip route
^
% Invalid input detected at '^' marker.

D1#show ip route
Codes: C - connected, S - static, ... (legend omitted)

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet1/1/1
L       192.168.10.1/32 is directly connected, GigabitEthernet1/1/1
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet1/1/2
L       192.168.11.1/32 is directly connected, GigabitEthernet1/1/2

```

Part 2: Compare a Layer 3 Switch and a Router

Step 1: Compare D1 and R1

Open the Physical tab on D1 and R1. Do you notice any similarities between the two? Do you notice any differences between the two?

Similarities: Both are rack-mounted devices that run Cisco IOS, both have a console port, and both accept SFP modules (GLC-LH-SMD appears in the module list of both).

Differences: D1 has 24 built-in Gigabit switchports plus 4 SFP uplink ports (28 total), while R1 has only 2 built-in Gigabit interfaces. R1 is modular: it has a NIM slot and its Modules list offers NIM-2T, NIM-ES2-4 and NIM-Cover, and a NIM-2T provides its 2 serial (WAN) interfaces. D1 has no serial ports, and its only modules are a power supply and SFPs.

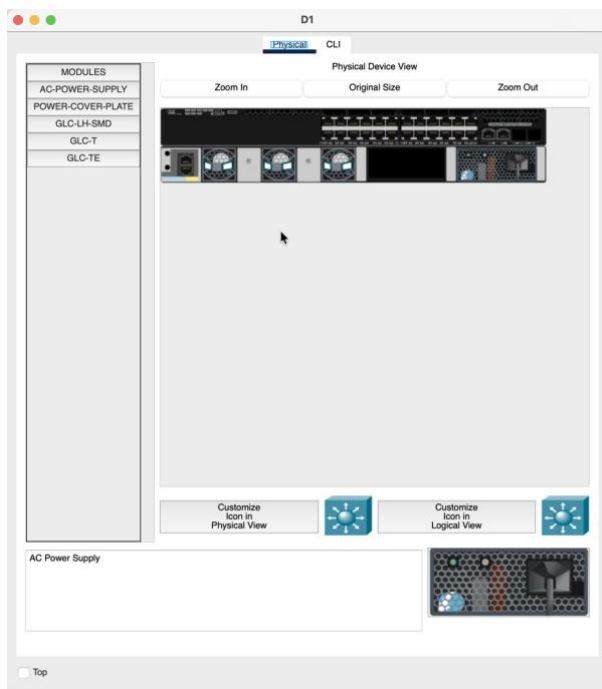


Figure 2a. Physical tab of D1 (3650-24PS)

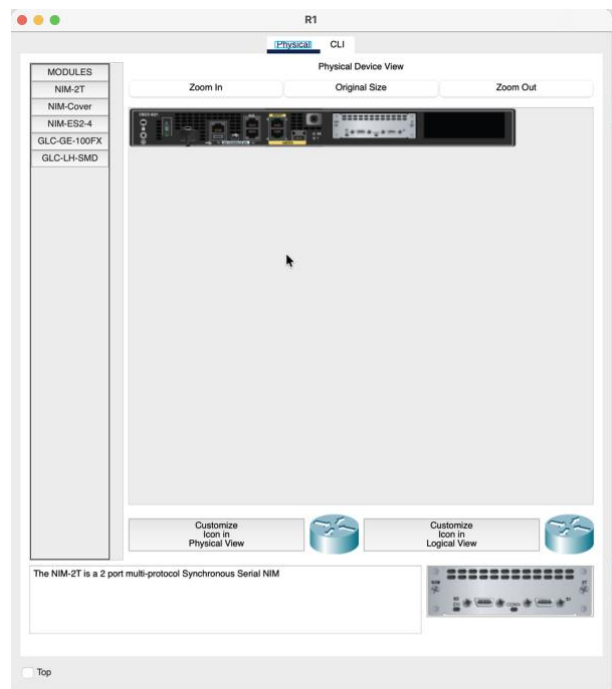


Figure 2b. Physical tab of R1 (4321) - NIM-2T module list

In the CLI tab, issue the show run command and examine the configurations of R1 and D1. What differences do you see between the two?

	R1 (4321 router)	D1 (3650 Layer 3 switch)
Setting an IP address	Configured directly on the port	Port must first be changed with no switchport
ip routing	Not shown (a router routes by default)	ip routing must be present
Interfaces	Gi0/0/0, Gi0/0/1, Serial0/1/0, Serial0/1/1	Gi1/0/1-24, Gi1/1/1-4 (no serial)
Serial interfaces	clock rate 2000000, shutdown	None
IOS version	15.4	16.3.2

Excerpt of show run on R1 (unrelated lines omitted).

```
R1#show run
...
interface GigabitEthernet0/0/0
 ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0/1
 ip address 192.168.11.1 255.255.255.0
!
interface Serial0/1/0
 no ip address
 clock rate 2000000
 shutdown
!
interface Serial0/1/1
 no ip address
 clock rate 2000000
 shutdown
```

Which command allows configuration of D1 with an IP address on one of its physical interfaces?
no switchport (entered in interface configuration mode). It changes the switch port from a Layer 2 port to a Layer 3 routed port, after which an IP address can be assigned.

Use the show ip route command on both devices. Do you see any similarities or differences between the two tables?

The two tables are essentially identical. Both show the directly connected networks 192.168.10.0/24 and 192.168.11.0/24 (C) with the local addresses (L), and neither has a gateway of last resort. The only difference is the interface names (Gi0/0/x on R1 and Gi1/1/x on D1). A Layer 3 switch with ip routing builds its routing table the same way a router does.

R1 output shown here; the D1 output is in Part 1 and has the same entries.

```
R1#show ip route
Codes: L - local, C - connected, S - static, ... (legend omitted)

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/0/1
```

Step 2: Compare R2 and D2

Analyze the routing table of R2 and D2. What is present now that was not present in the configuration of R1 and D1?

The tables now contain routes marked O (OSPF). These are learned dynamically from a neighbor through the OSPF routing protocol. The tables of R1 and D1 only had C (connected) and L (local)

routes. The OSPF adjacency message (from LOADING to FULL) in the console shows that R2 and D2 became OSPF neighbors and exchanged routes.

```
R2>en
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   1.1.1.0/24 is directly connected, GigabitEthernet0/0/1
   1.1.1.1/32 is directly connected, GigabitEthernet0/0/1
 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   10.11.48.0/30 is directly connected, GigabitEthernet0/0/0
   10.11.48.1/32 is directly connected, GigabitEthernet0/0/0
 O 192.168.10.0/24 [110/2] via 10.11.48.2, 00:07:20, GigabitEthernet0/0/0
 O 192.168.11.0/24 [110/2] via 10.11.48.2, 00:07:20, GigabitEthernet0/0/0

R2#
```

Figure 3a. show ip route on R2 - OSPF (O) routes to 192.168.10.0/24 and 192.168.11.0/24 via 10.11.48.2

```
D2>en
D2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 1.0.0.0/24 is subnetted, 1 subnets
   1.1.1.0 [110/2] via 10.11.48.1, 00:07:29, GigabitEthernet1/0/1
 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   10.11.48.0/30 is directly connected, GigabitEthernet1/0/1
   10.11.48.2/32 is directly connected, GigabitEthernet1/0/1
 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
   192.168.10.0/24 is directly connected, GigabitEthernet1/1/1
   192.168.10.1/32 is directly connected, GigabitEthernet1/1/1
 192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
   192.168.11.0/24 is directly connected, GigabitEthernet1/1/2
   192.168.11.1/32 is directly connected, GigabitEthernet1/1/2

D2#
```

Figure 3b. show ip route on D2 - OSPF (O) route to 1.1.1.0 via 10.11.48.1

Which network is in the routing table of D2 that was learned from R2?

1.1.1.0/24 - the entry O 1.1.1.0 [110/2] via 10.11.48.1, GigabitEthernet1/0/1 where 10.11.48.1 is the address of R2 on the link between R2 and D2.

Verify that each topology has full connectivity by pinging PC1 to PC2, PC3 to PC4, and PC5 to PC6 and PC7.

All pings succeeded. The first ping of each test had one "Request timed out" because of ARP resolution; the repeated ping received 4 of 4 replies (0% loss).

Test	Destination IP	Result (repeated ping)	Reply TTL
PC1 to PC2	192.168.11.10	4/4 replies, 0% loss	127
PC3 to PC4	192.168.11.10	4/4 replies, 0% loss	127
PC5 to PC6	192.168.10.10	4/4 replies, 0% loss	126
PC5 to PC7	192.168.11.10	4/4 replies, 0% loss	126

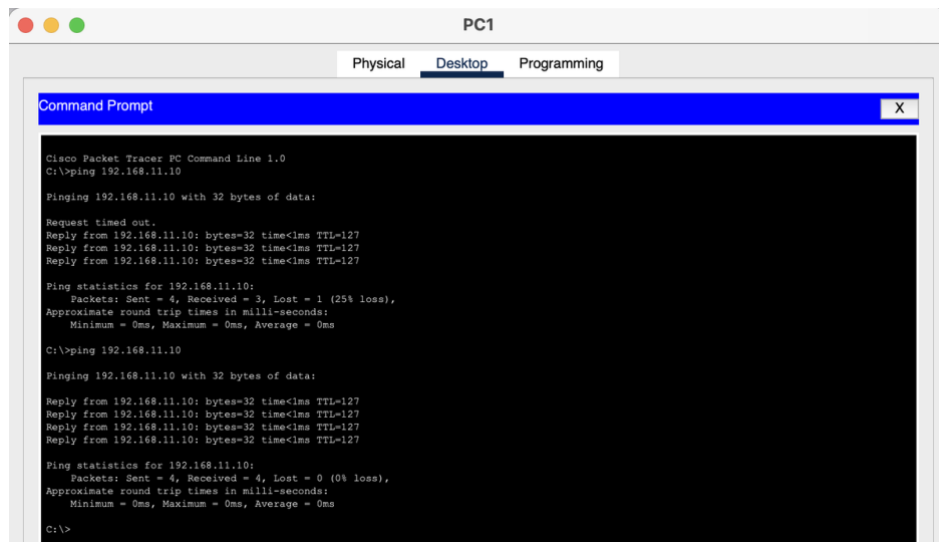


Figure 4a. Ping from PC1

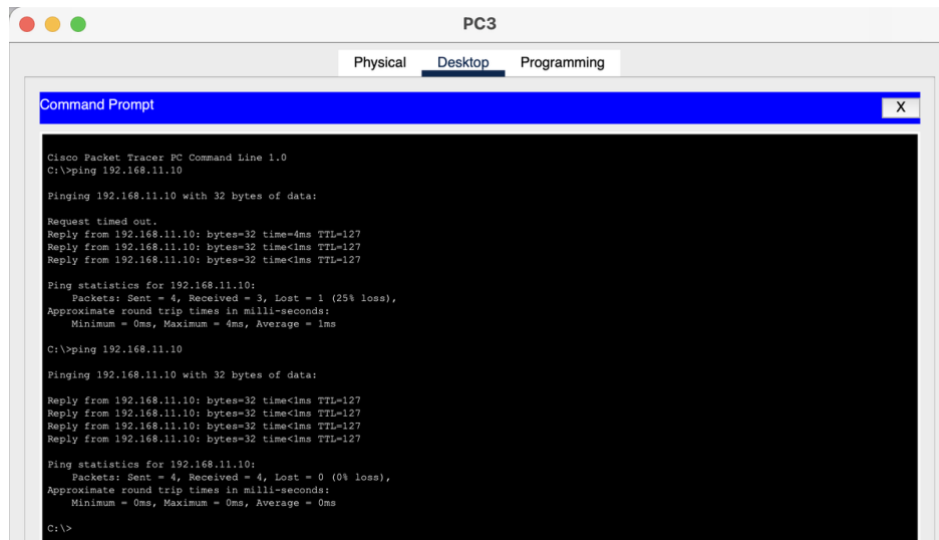
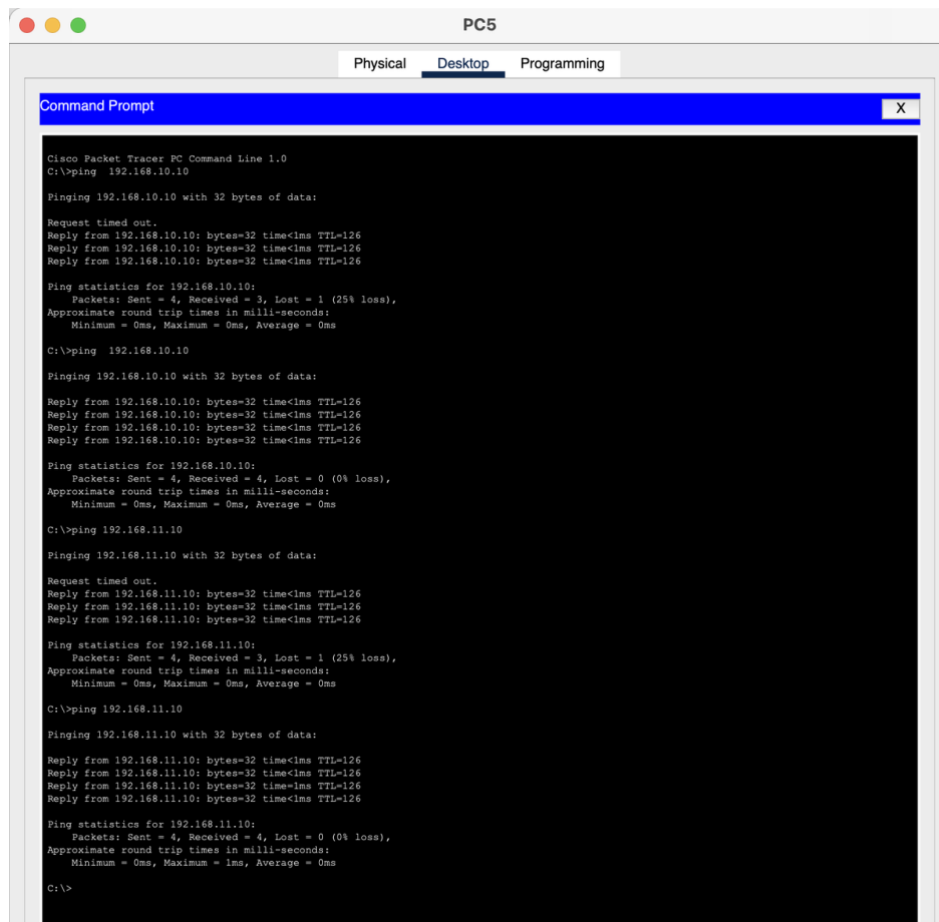


Figure 4b. Ping from PC3



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.11.10

Pinging 192.168.11.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.11.10

Pinging 192.168.11.10 with 32 bytes of data:

Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126
Reply from 192.168.11.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.11.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

Figure 4c. Ping from PC5 to both destinations

The reply TTL shows how many Layer 3 devices the packet passed through. A PC starts at TTL 128, and each Layer 3 hop lowers it by 1 (Layer 2 switches do not). TTL 127 means one Layer 3 device was crossed, and TTL 126 means two.

Which device is used to provide communication between networks?

A Layer 3 device is needed to provide communication between networks. This can be a router (R1, R2) or a Layer 3 switch with ip routing enabled (D1, D2).

Why were we able to ping across networks without there being a router?

In the first topology there is no router. D1 is a Layer 3 switch with ip routing enabled and routed ports with IP addresses (192.168.10.1 and 192.168.11.1) that act as the default gateways of the two networks. D1 builds a routing table with connected routes, as shown by show ip route, and forwards packets between the two networks just as a router would. The TTL of 127 in the replies shows that the packet crossed exactly one Layer 3 device.

Bonus question

Routers are Layer 3 devices and conventional switches are Layer 2 devices. However, we can assign an IP address to a management (SVI) interface of a Layer 2 switch. How is this possible if switches are Layer 2 devices?

The IP address on the SVI is only for managing the switch itself (for example Telnet/SSH access), so the switch acts like an ordinary host on that network. It uses the default gateway (ip default-gateway) to reach other networks. The switch still forwards frames by MAC address and does not route packets for other devices, because it has no ip routing and no routing table. This is why show ip route is not accepted on ASw-1, even though Vlan1 has the IP address 192.168.10.2.